

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2017- 2018 (Odd Semester)

Innovative Practices Description

UNIT IV (Conventional and Solid State Speed Control of D.C. Drives)

Degree, Semester & Branch: III Semester B.E. Mechanical Engineering A and B Section

Course Code & Title: EE6351 Electrical Drives and Control

Name of the Faculty member: Mr.N.Ganesh

Name of the Topic: Phase Controlled Rectifiers – Single Phase

Name of the Innovative Practice: Demonstration of Single Phase controlled rectifiers

Date & Duration: 12.09.2017 (50 Minutes) and 13.09.2017 (50 Minutes)

ICT Tool Used: Experimental Setup

Description: A practical demonstration session was conducted for students of II Mech – A and B on the topic titled, “Single phase controlled rectifiers”. The Students were explained about the construction, working and operation of single phase controlled rectifiers. The same was verified practically by conducting the experiment and explanation was given to student regarding variation of output voltage with respect to firing angle of the switches.

Goals (Learning Outcomes):

- The students will be able to explain the operation of Single Phase half controlled and fully controlled rectifier.
- The students will be able to compare Single Phase half controlled and fully controlled rectifier.

Use of appropriate methods:

Justification for Choosing the Activity for the topic:

- Differentiate Single Phase half controlled and fully controlled rectifier.

The topic Single Phase half controlled and fully controlled rectifier are used for supplying variable voltage to the dc motor. But Single Phase Semi converter has four switches of which, equal number of thyristors and diodes whereas, Single Phase fully controlled rectifier has four switches all of them are thyristors. After teaching the concept, I thought of conducting practical demonstration for making the students to give the difference between the two concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 50 Minutes

Effective presentation

Details of the Implementation:

After teaching the concept of Single Phase half controlled and fully controlled rectifier, the students were taken Power Electronics and Drives Laboratory.

Total Strength is 125 (61 + 64); Number of Sessions – 4;

Number of Students in Session 1 – 31; Number of Students in Session 2 - 30

Number of Students in Session 3 – 32; Number of Students in Session 4 - 32

Photographer: Lab Technician - Mr.S.Muthuraman

Reporter: Myself

At the end of the Unit IV (100 minutes - 50 minutes each for Section A and B)

Session 1 (25 minutes)

- ✓ I explained the students about the operation of Single Phase half controlled and fully controlled rectifier with necessary circuit diagram for 10 minutes.
- ✓ Then, I explained them how to connect the switches, triggering unit and load for another 5 minutes.
- ✓ Then, I demonstrated the operation of Single Phase Semi converter and fully controlled rectifier with various values of firing angle for another 5 minutes.
- ✓ Then, I asked the students to explain the operation of Single Phase Semi converter and fully controlled rectifier for another 3 minutes.
- ✓ Finally, I summarized the points told by the students.

Session 2 (25 minutes)

- ✓ I explained the students about the operation of Single Phase half controlled and fully controlled rectifier with necessary circuit diagram for 10 minutes.
- ✓ Then, I explained them how to connect the switches, triggering unit and load for another 5 minutes.
- ✓ Then, I demonstrated the operation of Single Phase half controlled and fully controlled rectifier with various values of firing angle for another 5 minutes.
- ✓ Then, I asked the students to explain the operation of Single Phase Semi converter and fully controlled rectifier for another 3 minutes.
- ✓ Finally, I summarized the points told by the students.

The same procedure is carried out for other two sessions (Session 3 and Session 4) for 25 minutes each.

A view of implementation of the Activity while the students are listening.



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when students told difference between Single Phase half controlled and fully controlled rectifier.
- ✓ While conducting the activity, I understood that the students can able to explain Single Phase half controlled and fully controlled rectifier without any confusion.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test III – Around 50% ((23+39) / (61+640)) of students answered correctly.**

Reflective Critique:

1. Pre-implementation Reflection :

- **Benefits:**

- Students can able to understand the concept by experimentally verifying with the theoretical concepts.
- Students can able to explain the concepts in examination without any confusion.

- **Challenges:**

- Since I had handled the subject for many times usually the students will confuse with the diagrams, formula and waveforms of Single Phase half controlled and fully controlled rectifier.
- In the class mostly boys hesitate to answer the questions.
- Time utilization for conducting activity.

2. Post-implementation Reflection :

- **Benefits:**

- Students understood the concept which was reflected from their answers for the questions I have asked during experimental demonstration.
- Students explained the concepts (theory and problems) in examination without any confusion over formula.

- **Challenges:**

- Students were formed in to two groups. Explanation was given to one group while the other group have waited for a long time in adjacent laboratory.
- One common setup was arranged to explain the concept for all the groups.

Changes required for future (if required):

- Several experimental setups can be arranged.
- Students can be allowed to do the experiments on their own and can clarify their doubts.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO3	2	2	2	1	1	-	-	-	-	-	-	1	2	2	1	-
CO4	3	2	2	2	1	-	-	-	-	-	-	1	2	2	1	-

References:

1. Vedam Subrahmaniam, "Electric Drives (concepts and applications)", Tata McGraw-Hill, 2001
2. Pillai.S.K "A first course on Electric drives", Wiley Eastern Limited, 1998
3. Singh. M.D., K.B.Khanchandani, "Power Electronics", Tata McGraw-Hill, 1998